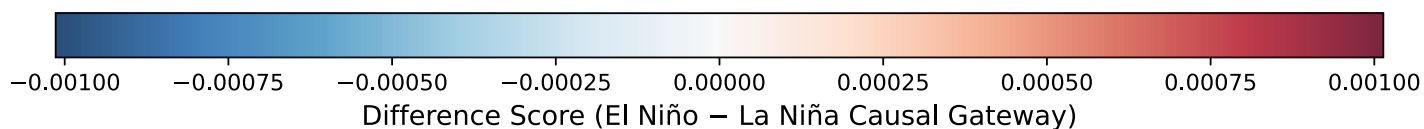
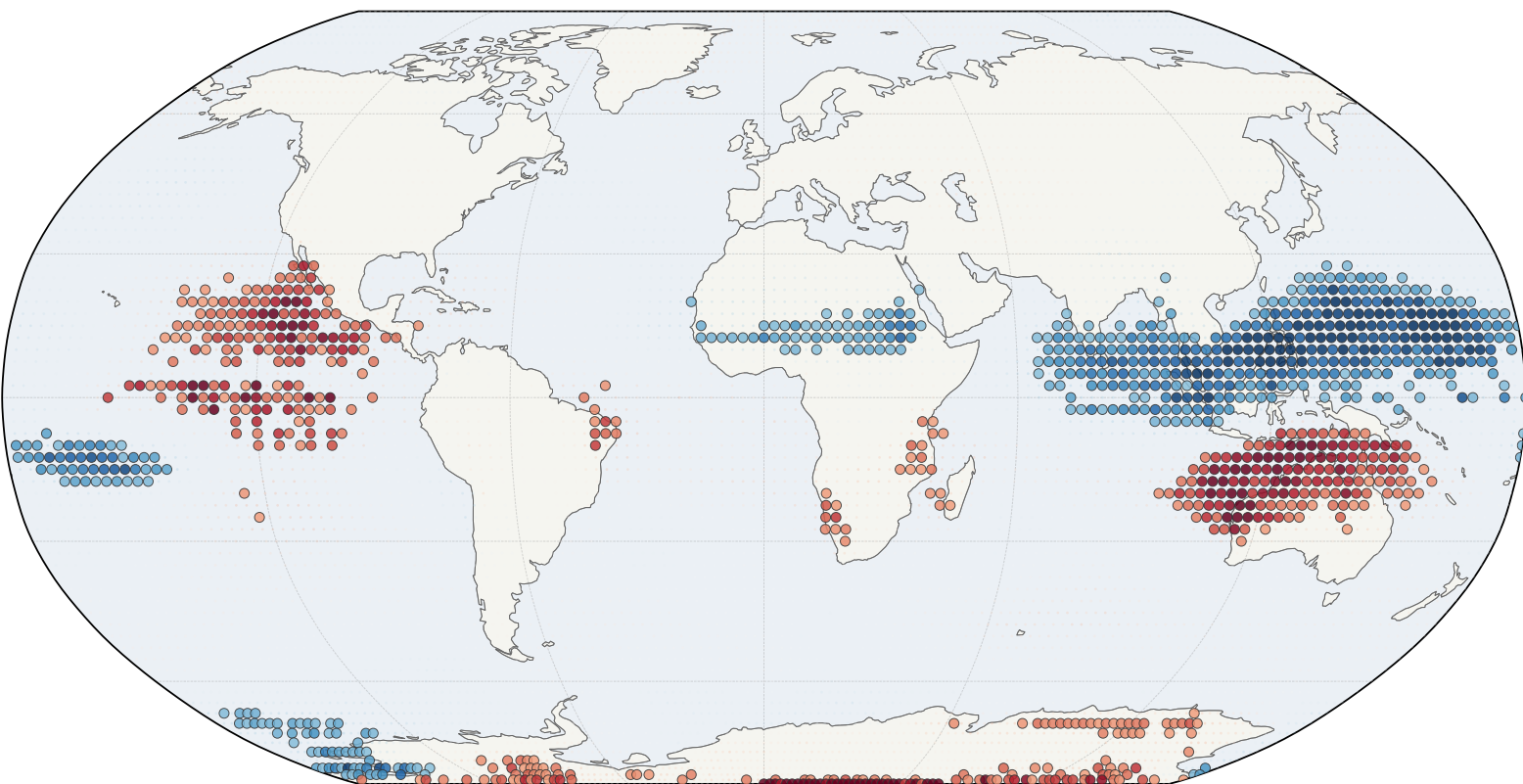
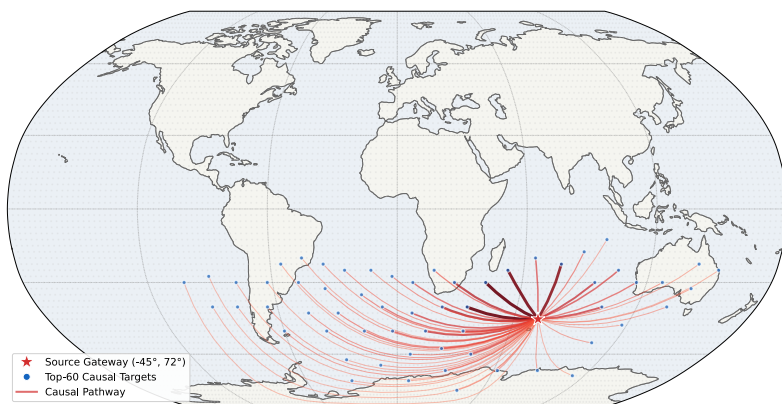


Topological Reorganization: El Niño vs La Niña (Winter NDJF)

Difference Map — Positive/Red = Stronger Causal Gateway in El Niño



Causal Flow Pathways from Top Gateway
(Line width $\propto$ causal strength)



Hierarchical Causal Structure
256 Clusters with Inter-Cluster Causal Flow (Top 1% edges)

